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The Temporal Moments of The Residential Green Environment Visits Associated with the Weather Observations

Abstract

In 17 EU Member States, the park is the most commonly cited setting for engaging in sports or other physical activities. However, there is limited data available on the use of residential green areas for recreational purposes and how this usage varies based on different weather conditions. Evidence of the weather on outdoor activities in residential green spaces is scarce. This study aims to explore the relationship between weather observations and visits to the nature reserve. Data includes two open data sets: a) the number of visits to the Lapakisto Nature Reserve in Lahti and b) the weather observations. The first data set was published by the City of Lahti, and the Finnish Meteorological Institute published the second. These two data sets are combined to investigate the association between the weather and the number of visits to the Lapakisto Nature Reserve. The results indicate how the number of visits associated with weather observations fluctuates monthly and between weekdays and weekends. The variation in visitor numbers in different weather observations means different needs for management and service actions produced by the public producers. Consequently, monitoring the number of park visitors, the use of the parks, and changes in the prevailing environmental and ecological values is important to safeguard the supreme quality of nature experiences available in smart and sustainable cities. In conclusion, weather diversity has both positive and negative effects on the use of recreational areas in cities.

Keywords: Residential green environments, open data, weather

1 Introduction

The use of parks is the most commonly reported setting for engaging in sports or other physical activities (PA) in EU Member States (European Commission, 2022, p.35). However, there is limited evidence on the use of residential green spaces for recreational purposes and how this varies under different meteorological conditions. Even though participation in outdoor PAs is affected by several factors that are challenging for an individual to control (Wagner et al. 2019), including changing seasons (Tucker & Gilliland 2007; Moschny et al. 2011), day length (Witham et al. 2014) and weather patterns (Tucker & Gilliland 2007; Moschny et al. 2011; Witham et al. 2014) such as temperature (Witham et al. 2014; Al-Mohannadi & Al-Kuwari 2015; Brocherie et al. 2015) wind intensity (Al-Mohannadi & Al-Kuwari 2015; Brocherie et al. 2015) and precipitation (Al-Mohannadi & Al-Kuwari 2015; Witham et al. 2014), data on the weather on outdoor activities in residential green spaces on a regional Lahti scale has not been studied yet. It has been shown by combining empirical estimates with outputs from downscaled climate models that unmitigated climate change will generate sizable activity-depressing and activity-delaying effects in summer when projected on an hourly resolution (Fan et al. 2023). In addition, most human-nature interaction research is conducted in spring and summer, with little insight into autumn or winter-human-nature interactions (Jakstis & Barnes 2024). Therefore, this study is justified, and it focuses on year-around statistics. We hypothesise that weather factors, which may interact with each other, are connected to the number of visits in city nature reserves by influencing an individual's perception of comfort in outdoor behaviour and the suitability of the green neighbourhood environment to outdoor activities.

The visits to nature parks and wildlife have been tracked with counters in Finland by Metsähallitus (Kajala & Erkkonen 2019), and the data on the visits is published yearly. In an international context, counters have been used to estimate the effect of weather on cycling in European cities (Goldmann & Wessel 2021) and the number of visits through the counter in urban American parks (Eco-Counter 2021). In addition, the National Outdoor Survey (Luonnonvarakeskus) tracks changes in outdoor recreation and many outdoor-related issues in people's daily lives. Data on the prevalence and frequency of outdoor recreation by Finns is collected every ten years.

In Finland, three cultural representations of outdoor activities in nature have been identified and conceptualised (Simula 2012). Traditional-pragmatic representation connects the relationship of rural-culture heritage with nature to the contemporary interpretations of outdoor activities. It reflects the meanings of traditional practices of outdoor activities in nature. Romantic-expressive representation stems from the conception of nature created by the romantic movement. The meanings of outdoor activities in nature are attached to nature experience and the mind's release of everyday thinking. Recreational-collective representation is connected with the people's movement of nature-goers and recreational cultures. Nature is formed to be a recreational environment, and outdoor activities in nature as a recreational-collective content of life (Simula 2012).

Almost 90% of Finns consider nature essential or very important (Ranta & Ahtiainen 2022). There are some differences in the importance of nature by age group. Finns' most important natural areas are green areas and parks near the home. The most popular activities are relaxing and exercising in nature.

This case study aims to explore the temporal moments of residential green environment visits in Lapakisto Nature Reserve, Lahti, Finland, associated with local weather observations: air temperature, precipitation and snow depth. Understanding the relationship between weather observations and the number of park visits will help the park management establish resources to balance the supply of park services with the demand based on the predicted volume of the park visits.

2 Methods

This study includes two open data sets: a) the number of visits to the Lapakisto Nature Reserve Park in Lahti recorded by a counter (#12 Luontopolku) located at the starting point of nature paths (see Figure 1 to see the location of the counter), and b) the daily weather data (sum of precipitation, average air temperature, and snow depth at 01:00 UTC) of the nearest observation station (Sopenkorpi, Lahti Finland). The first data set was published by Eco-Counter® France/World (Eco-Counter), and the second by the Finnish Meteorological Institute (Ilmatieteen laitos). The counter is used as a measuring instrument to detect pedestrians passing by and record the number of passers-by (referred to in the text as visitors) to the online database, from which the Lapakisto data is downloaded via API for the current study. The distance from Lapakisto to the observation station is 13 kilometres. We assume that the majority of the visitors arrive in Lapakisto from Lahti. Therefore, the observation station with the closest proximity to the origin of the visitors will give the best information about the weather parameters related to the visitor's planned behaviour for a visit to Lapakisto. We used the counter data for the maximum available time from August 2018 until April 2024. The counter data was initially recorded at a more granular level (every 15 minutes) and had to be aggregated to calculate the daily sum of visitors. We downloaded the weather data from the open data service (Ilmatieteen laitos) to get the data for the same time interval. Then we used the dplyr package (Wickham et al. 2023) in R and the date information of the data to join the two data sets together to investigate the associations between the weather (temperature, precipitation, and snow depth) and the number of visits to Lapakisto. We used daily data and lubridate (Grolemund & Wickham 2011) to aggregate new variables to show a) which day of the week visits had taken place and then transformed this data into a variable showing if the visits happened during weekday or weekend, and b) we extracted the month from data to code each visit into a particular month. With the help of aggregated variables, the combined data determines the temporal moments relating to the number of visits. We conducted a correlation analysis (Pearson's r) in Jamovi (2024) to inspect the relationship between visits and weather observations.

2.1 Lapakisto Nature Reserve Park

Lapakisto Nature Reserve Park is a diverse 230-hectare protected area of lakes, rocky outcrops and small bogs 11 kilometres from the city centre of Lahti (Lahden seudun luonto). There are three ponds within. About a quarter of the forests in the area are old-growth forests, and swamps cover about 15% of the area and lakes about 10%. There are five campfire sites and one camping hut. The Lahti

city provides firewood at all campfire sites. The area is crisscrossed by five marked trails totalling 23 km in length. A pedestrian counter is located near the site's parking area, from which an accessible route to the campfire sites starts.

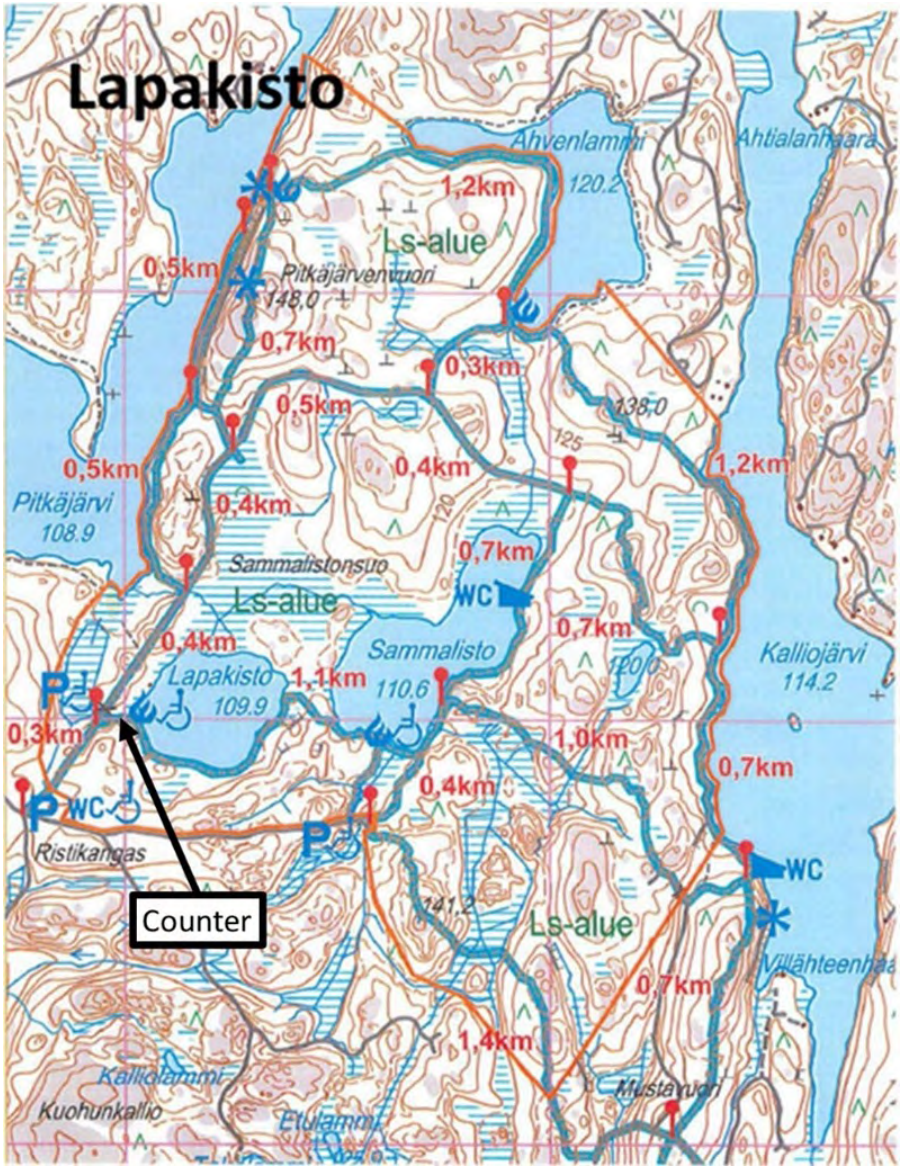


Figure 1. The Lapakisto Nature Reserve Park map in Lahti, Finland, has marked trails and a counter. (Maanmittauslaitos)

3 Results and discussion

During the observation period, the counter recorded 283600 visits to the Lapakisto nature reserve. The mean number of visits per calendar year has been 50730, while the maximum amount was 68591 in 2020 during the COVID-19 pandemic, turning to an annual decline, reaching the minimum (40899) in 2023.

3.1 The number of visits

The number of visits fluctuates monthly and between weekdays and weekends (Figure 2). The visits took place mainly during the weekends rather than the weekdays, except for the most common holiday season (June and July) in Finland when the difference between visits during weekdays and weekends is smaller. The number of weekend visits is the highest in autumn (from August to November). Then, during the darkest months (from December to February), visits decreased before the incline during late winter/early spring (March and April).

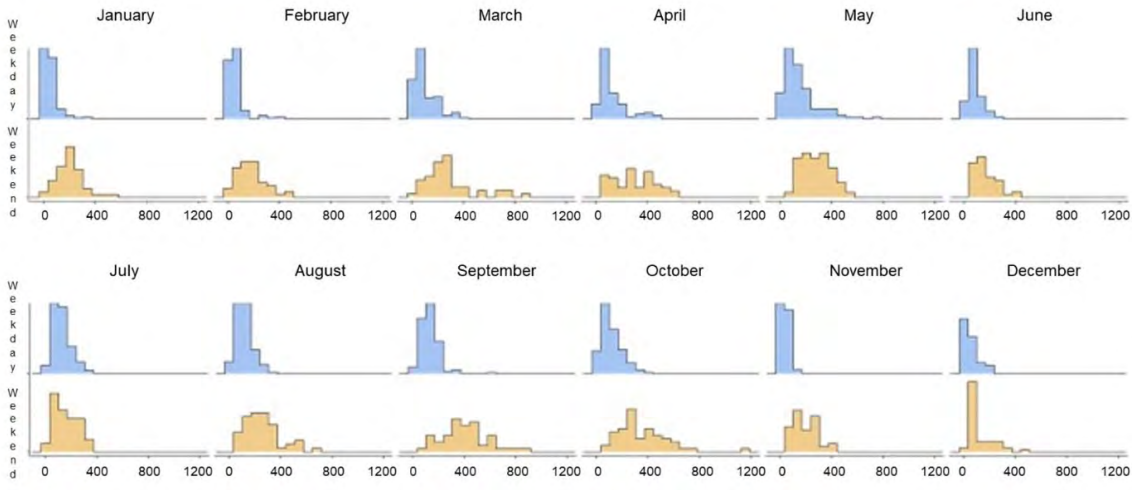


Figure 2. Histogram of visits through the counter at Lapakisto Nature Reserve Park per month divided into weekdays and weekends.

The detailed information about the visits to Lapakisto is presented in Table 1. The first column shows the month (1 = January, 2 = February, etc.), for which the next column divides the observations (days) into weekdays and weekends, of which the number of observations is then presented with the following columns displaying the descriptive information of the visits.

3.2 Weather data

The weather data consists of a wide variation of precipitation, temperature and snow depth during the observation period. For precipitation, 2060 observations were made, with six missing data points. The mean precipitation was 1.69 mm (sd 3.7 mm), with a median of 0.10 mm. The minimum recorded precipitation was 0.0 mm, while the maximum was 62.4 mm. Regarding temperature, there were 2058 observations, with eight missing data points. The mean temperature was 5.63°C, with a median of 4.70°C (sd 9.3°C). The minimum recorded temperature was -24.2°C, while the maximum was 26.9°C. 1881 observations were made regarding snow depth, with 185 missing data points. The mean snow depth was 8.84 cm (sd 15.5 cm), with a median of 0.0 cm. The minimum recorded snow depth was 0.0 cm, while the maximum was 65.0 cm.

Table 1. The number of visitors.

Month		N	Missing	Mean	Median	SD	Minimum	Maximum	Skewness	SE	Kurtosis	SE	p
1	weekday	133	0	52.0	36	59.0	2	373	2.935	0.210	10.799	0.417	< .001
	weekend	53	0	204.7	196	106.7	22	555	0.988	0.327	1.916	0.644	.011
2	weekday	121	0	59.0	41	68.1	1	376	3.288	0.220	11.786	0.437	< .001
	weekend	49	0	180.5	170	109.1	25	469	0.779	0.340	0.188	0.668	.019
3	weekday	133	0	97.7	73	82.9	0	381	1.331	0.210	1.340	0.417	< .001
	weekend	47	6	287.1	246	202.2	28	911	1.440	0.347	1.710	0.681	< .001
4	weekday	116	0	124.7	84.5	109.0	2	477	1.701	0.225	2.315	0.446	< .001
	weekend	45	0	298.0	289	155.7	42	601	0.157	0.354	-0.957	0.695	.145
5	weekday	110	0	157.3	112.5	136.6	12	757	1.864	0.230	4.038	0.457	< .001
	weekend	45	0	283.6	272	112.5	82	518	0.330	0.354	-0.656	0.695	.311
6	weekday	108	0	86.9	79.0	54.1	7	300	1.287	0.233	2.129	0.461	< .001
	weekend	42	0	169.5	157.5	87.6	38	423	1.028	0.365	0.884	0.717	.006
7	weekday	110	0	127.5	115.0	68.0	15	313	0.797	0.230	0.250	0.457	< .001
	weekend	45	0	161.0	148	90.2	10	369	0.518	0.354	-0.611	0.695	.080
8	weekday	123	0	119.7	117	63.2	4	325	0.740	0.218	0.585	0.433	.002
	weekend	48	0	260.3	236.5	140.1	51	686	0.938	0.343	0.763	0.674	.012
9	weekday	128	0	131.8	121.0	80.2	2	626	2.252	0.214	10.950	0.425	< .001
	weekend	52	0	402.7	413.0	177.2	66	879	0.472	0.330	0.354	0.650	.345
10	weekday	132	0	118.1	97.0	80.6	1	431	1.100	0.211	1.228	0.419	< .001
	weekend	54	0	349.9	292.0	195.7	43	1157	1.455	0.325	4.137	0.639	< .001
11	weekday	130	0	37.1	32.0	26.2	0	138	1.281	0.212	2.250	0.422	< .001
	weekend	50	0	192.8	190.0	91.7	58	432	0.468	0.337	-0.147	0.662	.054
12	weekday	132	0	55.6	35.5	54.4	0	231	1.447	0.211	1.307	0.419	< .001
	weekend	54	0	117.5	78.5	95.7	9	479	1.673	0.325	2.820	0.639	< .001

3.3 Relation between visits and weather observations

A weak positive correlation was found between the number of visits and the temperature $r(2050) = .18$, $p < .001$. When the relation between the observed variables is investigated visually (Figure 3), it is seen that the relation between the number of visits and the temperature is not that straightforward. It seems that the number of visits increases when the temperature rises until around 10°C, and when the temperature rises beyond that point, the number of visits declines during the warmest days in June and July. On the other hand, snow depth has only a weak negative correlation $r(1873) = -.18$, $p < .001$ with the number of visits. In addition, precipitation has a weak negative correlation $r(2052) = -.17$, $p < .001$ with the number of visitors.

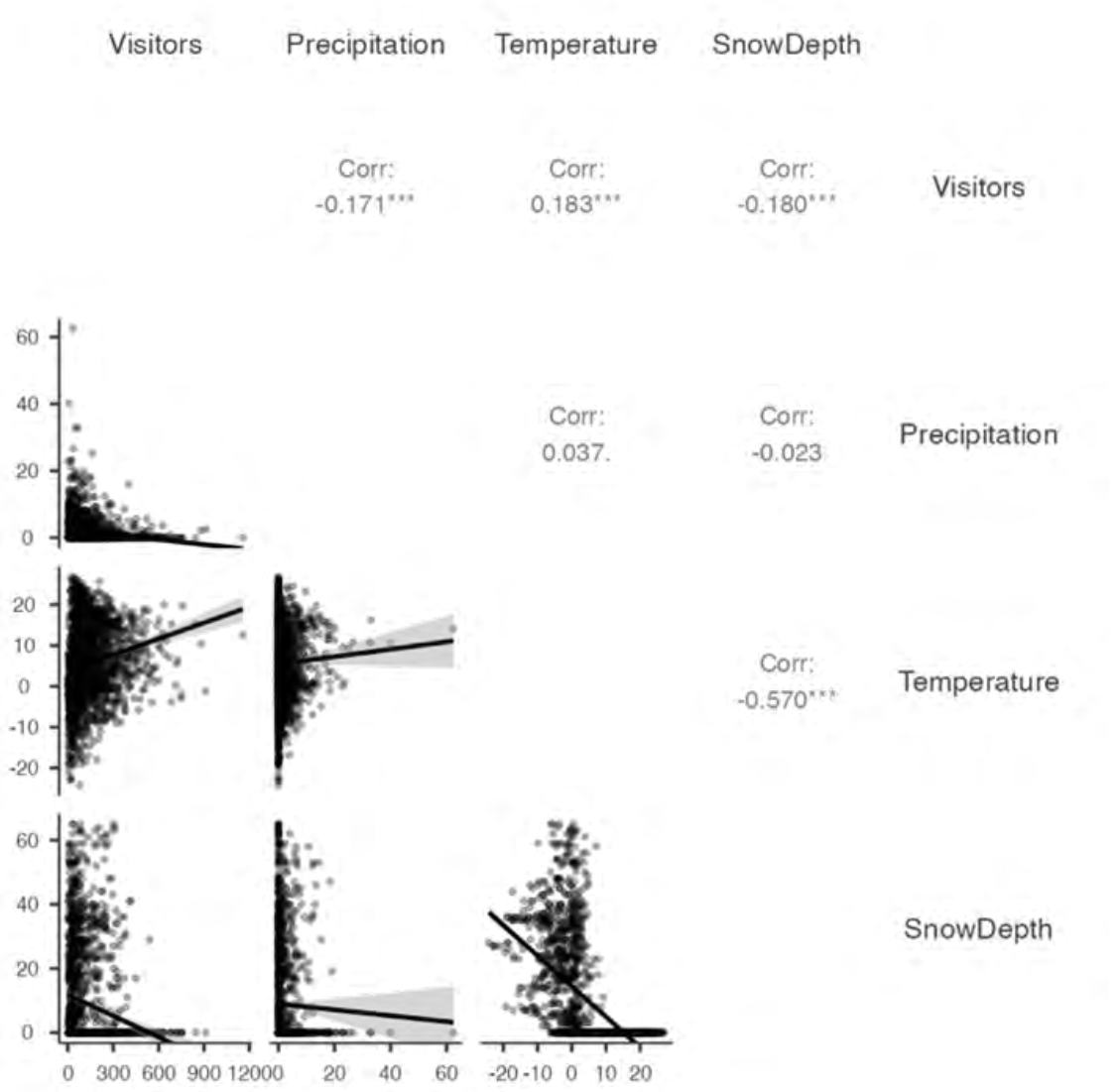


Figure 3. Scatterplot Matrix of the number of Lapakisto Nature Reserve Park visitors per day (n=2060) and weather observations: precipitation (n=2052), temperature (n=2050), and snow depth (n=1873) .

There are some limitations to the study. The weather data was collected from a location different from the exact location of the counter. Thus, the weather at the counter can be good, but the observation station data does not recommend a park visit. The decisions for the visits observed in the study are likely made well before the visits. Therefore, the current weather observation is not likely the best predictor for a park visit. Perhaps the weather observation an hour before or from the same morning might have a stronger correlation with the number of visits. The relationship between these factors should be studied in the future.

4 Conclusions

This case study aimed to explore the relationship between residential green environment visits in Lapakisto Nature Reserve and local weather observations. In general, based on this study's results, people visit nature reserve more often on weekends than on weekdays. This is a similar finding to the previous data gathered from urban parks in the Los Angeles region, but there was a seasonal difference between them (Eco-Counter 2021). In Lahti, autumn was the annual peak for visitors. Contrary to our hypothesis, neither the rain nor snow seemed to be a significant obstacle to Lapakisto nature reserve visits. In conclusion, based on the results of this study, the positive or negative effects of weather diversity on visits to recreational areas in Lahti cannot be stated precisely. These findings shed more light on the relationship between weather observations and recreational outdoor activity levels (Witham et al. 2014; Al-Mohannadi & Al-Kuwari 2015). In addition, this study expands the understanding of the weather observations with the role of snow depth as a novel contribution at the regional level in Finland.

In this study, each current variant typically provides only a small fraction of the total. Further studies would be needed to count the overall risk of polygenic variants to mathematically model how the contribution of all these variants, when added together, would give a combined risk value for their combined effect on inactivity.

The possibility of using city nature for recreation is highly important in Finland. However, it is not self-evident that this strong tradition will continue. In future, trends such as climate change, urbanisation, ageing and immigration will pose challenges to these outdoor activities. Consequently, monitoring the number of park visitors, the use of the parks, and changes in the prevailing environmental and ecological values is important to safeguard the supreme quality of nature experiences available in smart and sustainable cities. Considering the practical implications of the current study, these findings can help understand the behaviour of park visitors. When the fluctuation of visits is predicted, it becomes easier to plan the park services. E.g., firewood replenishments, emptying the bins, transport services, and other maintenance activities can be planned following the patterns of visits. These findings call for more attention to leveraging real-time activity data to understand intraday adaptation behaviours and their associated behavioural consequences in outdoor activity studies.

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